# Walid Irfan

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## ABOUT ME

Electrical Power Systems Engineer with 7 years of experience across semiconductor and critical infrastructure projects. Currently serving as the Owner's Engineer representative for the €1.3B STMicroelectronics 300mm semiconductor fab expansion in Italy. Expertise in HV/MV/LV power system design, equipment sizing, protection & control, commissioning, preventive maintenance, and reliability improvement. Proficient in power system studies including load flow, short-circuit, protection coordination, transient stability, and transformer diagnostics using ETAP. Experienced with 132kV/11kV substations, relay commissioning, diesel generator systems, and BIM coordination within multidisciplinary environments.

## WORK EXPERIENCE

### ELECTRICAL CONSTRUCTION MANAGER – ARTELIA ITALIA – 15/12/2025 – Current – MILAN, ITALY

- Acting as Owner's Engineer representative on the **LV/MV electrical scope** of one of Europe's landmark semiconductor manufacturing project, a **€1.3 billion facility expansion** in Northern Italy, representing a **flagship project** in European semiconductor sovereignty and **advanced manufacturing infrastructure**.
- **Managed** consultants, subcontractors, vendors, and site teams to **ensure timely execution** of **electrical works** in line with **project schedules**, **design requirements**, and **construction milestones**.
- **Managed project progress** reviews and coordinated with project stakeholders to resolve technical and execution-related issues while maintaining **alignment with project timelines.**
- Prepared RFQs, technical specifications, and engineering documentation defining **project requirements**, **applicable IEC/CEI standards**, and **technical performance criteria** for **contractors and vendors**.
- **Supervised commissioning activities** including FAT, SAT, pre-commissioning inspections, system readiness verification, and validation of protection settings prior to energization **in accordance with IEC 61439.**
- **Reviewed and validated** LV/MV electrical designs, load calculations, cable sizing, voltage drop, short-circuit analysis, and protection coordination studies using ETAP, Ampere, and Caneco BT in accordance with **IEC 60364 and CEI 64-8.**
- Supported **technical evaluation** and comparison of **vendor proposals,** preparing **compliance assessments** and **technical reports** to **optimize cost**, **performance**, and **equipment selection**.

### PAKISTAN ATOMIC ENERGY COMMISSION – KARACHI, PAKISTAN

#### SENIOR ELECTRICAL ENGINEER – 01/12/2023 – 30/08/2024

- **Led a team of electrical engineers and technicians** in the maintenance and troubleshooting of **rotating equipment, power distribution systems, and industrial machinery**, ensuring minimal downtime and optimal equipment performance.
- **Ensured compliance with HSE guidelines and international safety standards**, conducting **risk assessments, hazard identification**, and implementing **preventive safety measures** across electrical operations.
- **Managed 132/11kV feeder operations**, including **annual maintenance scheduling, relay testing**, and recommissioning of **HV protection systems**, ensuring system reliability and regulatory compliance.
- **Supervised and optimized the performance** of **810kVA diesel generators** equipped with **DSE-8610 controllers**, enabling seamless **Auto Transfer Switching (ATS)** and uninterrupted power supply during grid outages.
- **Conducted Factory Acceptance Tests (FAT), Site Acceptance Tests (SAT), and commissioning** of electrical equipment, verifying compliance with **OEM specifications, project standards**, and functional performance criteria.

#### JUNIOR ELECTRICAL ENGINEER – 30/12/2021 – 30/11/2023

- **Troubleshot and maintained** underground power distribution systems, high-voltage panels (up to 11kV), and transformers, helping to reduce power restoration time by **20%**.
- **Performed Root Cause Analysis (RCA)** and predictive maintenance practices, achieving **15% decrease** in unexpected equipment failures.

- **Maintained and calibrated** industrial instrumentation to ensure accurate process control and minimize downtime.
- **Designed a 3MW power distribution and illumination system** using **ETAP**, **AutoCAD**, and **EPLAN**.with voltage drops kept within **5%** of nominal levels.
- **Led site supervision and execution** of a 3MW electrical distribution system, ensuring proper cable sizing, panel selection, and lighting layouts..
- **Installed and configured 810kVA Diesel Generators**, integrating **DSE-8610 and DSE-8660 controllers** for Auto Transfer Switching (ATS), which cut switchover time during mains failure by **over 50%**.

### GRADUATE ELECTRICAL ENGINEER (FELLOWSHIP PROGRAM) – 12/2019 – 12/2021

- Completed a two-year, industry-integrated **fellowship program** focused on **power generation, transmission,** and **system reliability**.
- Collaborated daily with engineers at an **operational power plant**, for on-site electrical and mechanical **challenges and regulatory practices**.
- Learned directly from **industry professionals** through applied courses in **power systems**, **protection**, and **electrical design**.
- Performed **laboratory simulations** and **prototype testing** replicating real-world **power plant systems** and **grid interconnections**.
- Utilized **ETAP software** to model power plant networks & perform **load flow**, **short circuit**, and **transient stability studies**.
- Collaborated with the **Pakistan National Regulatory Authority** on research related to **grid integration**, **compliance**, and **energy standards**.

### PROJECT INTERN – K-ELECTRIC (ELECTRIC UTILITY COMPANY) – 09/2018 – 08/2019 – KARACHI, PAKISTAN

- **Transformer diagnostics** using **FMEA** and **preventive maintenance** to improve **reliability** and **reduce downtime**.
- Developed a **MATLAB**-based tool for **data-driven condition monitoring** of transformer **health index** & **lifespan estimation.**
- Gained experience in **MV/LV switchgear, HV protection**, and **substation automation** (fault analysis, load flow, protection coordination).

### ENGINEERING INTERN – SCHLUMBERGER – 06/2018 – 07/2018 – KARACHI, PAKISTAN

- Adapted to field environments, applying **problem-solving** skills to support **maintenance and testing** of drilling equipment and **electrical systems**.
- Collaborated with **multicultural teams** (engineers, technicians, vendors), ensuring **effective communication** and coordination in operations.

## EDUCATION AND TRAINING

09/2024 – CURRENT Milan, Italy

**MS IN ELECTRICAL ENGINEERING** Politecnico di Milano

**Website** https://www.polimi.it/

26/12/2015 – 20/07/2019 Karachi, Pakistan

**B.E. IN ELECTRICAL ENGINEERING** NED University of Engineering & Technology.

**Website** https://www.neduet.edu.pk/ | **Final grade** 3.79/4.0

## PUBLICATIONS

2022

**N-1 Contingency Analysis for Offsite Power System of an HPR-1000 Power Plant Using ETAP Software**

2022

**Transient Stability Analysis of an HPR-1000 Power Plant using ETAP Software**